

October 28, 2016

Ms. Amy Vail
City of Cadillac WWTP
200 N. Lake St.
Cadillac, MI 49601

Subject: Executive Summary for Project Number 211750

Dear Ms. Vail:

Please find the attached biomonitoring report for testing conducted October 14, 2016 through October 21, 2016. The results for this evaluation are:

Test Species	TUa	TUc	MATC
Ceriodaphnia dubia	0	0	>100
Fathead Minnows	0	0	>100

The exposure tests were valid since they met the criteria for survival, reproduction, and growth in the controls. Should you have any questions pertaining to this report please let me know.

Sincerely,
Michael Amboyan
Paragon Laboratories, Inc.

DEQ
MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY - WATER BUREAU
CERIODAPHNIA DUBIA CHRONIC TOXICITY TEST REPORT
By authority of PA 451 of 1994, as amended.

1. NAME OF FACILITY (on NPDES permit) Cadillac WWTP				2. NPDES PERMIT # MI0020257			
3. RECEIVING WATER (as designated in permit) Saline River			4. OUTFALL 001		5. RECEIVING WATER CONCENTRATION (if known) Unknown-Cerio + FH		
6. TEST LAB (Name and Address) Paragon Laboratories, Inc. 12649 Richfield Ct. Livonia MI 48150							
7. TEST START DATE 10/14/16		8. TEST END DATE 10/20/16		9. AGE RANGE OF ORGANISMS AT TEST START Less Than 24 Hours		10. Report Date 10/28/2016 9:19:06 AM	
11. NAME OF PERSON CONDUCTING TEST Michael Amboyan				12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Paragon Laboratories, Inc. (734) 462-3900			
13. SAMPLE COLLECTION DATES 10/13/2016 7:00:00 AM		14. DATE RECEIVED 10/14/2016 10:16:00 AM		15. DATE OF FIRST USE 10/14/2016		16. ARRIVAL TEMP (°C) Sample 1: 4 on ice	
10/17/2016 7:00:00 AM		10/18/2016 9:56:00 AM		10/18/2016		Sample 2: 4 on ice	
10/18/2016 7:00:00 AM		10/19/2016 9:50:00 AM		10/19/2016		Sample 3: 5 on ice	
17. TOTAL RESIDUAL CHLORINE Sample 1: <0.02		18. AMMONIA (mg/L as N) Sample 1: <0.5		19. pH Sample 1: 7.53		20. Alkalinity Sample 1: 107.2	
Sample 2: <0.02		Sample 2: <0.5		Sample 2: 7.67		Sample 2: 97.7	
Sample 3: <0.02		Sample 3: <0.5		Sample 3: 7.77		Sample 3: 100.4	
21. Hardness Sample 1: 211.9							
Sample 2: 185.4							
Sample 3: 193.1							
22. WAS SAMPLE DECHLORINATED? Sample 1: No Sample 2: No Sample 3: No		23. DESCRIBE DECHLORINATION (if any) N/A					
24. EFFLUENT SAMPLES WERE COLLECTED: Facility Does Not Chlorinate							
25. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded hold time.) N/A							
26. EFFLUENT FILTERED? No		27. STATE MESH SIZE OF FILTER (if filtered) N/A					
28. EFFLUENT SAMPLE TYPE: Sample 1: 24-Hour Composite Sample 2: 24-Hour Composite Sample 3: 24-Hour Composite						29. IDENTIFY THE DILUENT (O1) MHW IDENTIFY THE SECONDARY (O2) N/A	
30. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCITON							
CONCENTRATION OF EFFLUENT (%)	O1	5	15	30	60	100	
48-HOUR SURVIVAL (%)	100	100	100	100	100	100	
6-Day MEAN REPRODUCTION/FEMALE	21.8	20.5	24.8	21.9	23.5	23.8	
6-Day MEAN SURVIVAL (%)	100.0	90.0	100.0	90.0	100.0	100.0	
31. 48-HOUR LC50 (%) >100		32. TUa (acute toxic units) 0					
33. 6-Day CHRONIC VALUE (%) >100		34. NOEC 100		35. LOEC >100		36. TUc (chronic toxic units) 0	

Prepared by Paragon Laboratories, Inc. Livonia MI

FATHEAD MINNOW CHRONIC TOXICITY TEST REPORT

By authority of PA 451 of 1994, as amended.

1. NAME OF FACILITY (on NPDES permit) Cadillac WWTP				2. NPDES PERMIT # MI0020257			
3. RECEIVING WATER (as designated in permit) Saline River			4. OUTFALL 001		5. RECEIVING WATER CONCENTRATION (if known) Unknown-Cerio + FH		
6. TEST LAB (Name and Address) Paragon Laboratories, Inc. 12649 Richfield Ct. Livonia MI 48150							
7. TEST START DATE 10/14/16		8. TEST END DATE 10/21/16		9. AGE RANGE OF ORGANISMS AT TEST START Less Than 24 Hours		10. Report Date 10/28/2016 9:19:06 AM	
11. NAME OF PERSON CONDUCTING TEST Michael Amboyan				12. NAME/PHONE # OF PERSON WHO CAN ANSWER QUESTIONS ABOUT THIS REPORT Paragon Laboratories, Inc. (734) 462-3900			
13. SAMPLE COLLECTION DATES 10/13/2016 7:00:00 AM		14. DATE RECEIVED 10/14/2016 10:16:00 AM		15. DATE OF FIRST USE 10/14/2016		16. ARRIVAL TEMP (°C) Sample 1: 4 on ice	
10/17/2016 7:00:00 AM		10/18/2016 9:56:00 AM		10/18/2016		Sample 2: 4 on ice	
10/18/2016 7:00:00 AM		10/19/2016 9:50:00 AM		10/19/2016		Sample 3: 5 on ice	
17. TOTAL RESIDUAL CHLORINE Sample 1: <0.02		18. AMMONIA (mg/L as N) Sample 1: <0.5		19. pH Sample 1: 7.53		20. Alkalinity Sample 1: 107.2	
Sample 2: <0.02		Sample 2: <0.5		Sample 2: 7.67		Sample 2: 97.7	
Sample 3: <0.02		Sample 3: <0.5		Sample 3: 7.77		Sample 3: 100.4	
21. Hardness Sample 1: 211.9		Sample 2: 185.4		Sample 3: 193.1			
22. WAS SAMPLE DECHLORINATED? Sample 1: No Sample 2: No Sample 3: No		23. DESCRIBE DECHLORINATION (if any) N/A					
24. EFFLUENT SAMPLES WERE COLLECTED: Facility Does Not Chlorinate							
25. DESCRIBE ANY DEVIATIONS FROM TEST METHODS (For example, pH-controlled test, reduced DO levels in test leading to aeration, sample exceeded hold time.) N/A							
26. EFFLUENT FILTERED? No		27. STATE MESH SIZE OF FILTER (if filtered) N/A					
28. EFFLUENT SAMPLE TYPE: Sample 1: 24-Hour Composite Sample 2: 24-Hour Composite Sample 3: 24-Hour Composite						29. IDENTIFY THE DILUENT (O1) MHW IDENTIFY THE SECONDARY (O2) N/A	
30. SUMMARY OF DATA AND RESULTS - SURVIVAL AND REPRODUCTON							
CONCENTRATION OF EFFLUENT (%)	O1	5	15	30	60	100	
96-HOUR SURVIVAL (%)	100	97.5	92.5	90	95	95	
7-DAY MEAN BIOMASS (mg/initial fish)	0.2683	0.2665	0.2588	0.2410	0.2850	0.3395	
7-DAY MEAN SURVIVAL (%)	95	95	90	87.5	92.5	92.5	
31. 96-HOUR LC50 (%) >100		32. TUa (acute toxic units) 0					
33. 7-DAY CHRONIC VALUE (%) >100		34. NOEC 100		35. LOEC >100		36. TUc (chronic toxic units) 0	

TOXICITY EVALUATION
Ceriodaphnia Dubia Summary Table

Client: City of Cadillac WWTP
 Project Number: 211750
 Primary Analyst: Michael Amboyan

Effluent Concentration	TOTAL LIVE YOUNG IN FIRST 3 BROODS PER REPLICATE										Total No. of Young	Total No. of Adults
	Replicate Number											
	1	2	3	4	5	6	7	8	9	10		
0%	24	23	26	29	28	13	25	5	28	17	218	10
5%	17	30	28	31	28	18	27	0	13	13	205	9
								A				
15%	27	31	28	14	32	17	29	23	30	17	248	10
30%	14	28	30	31	32	15	27	13	29	0	219	9
										A		
60%	24	27	21	26	35	24	28	16	18	16	235	10
100%	28	29	27	30	29	11	13	27	28	16	238	10

Qualifier Key:

A = Natural Death
 B = Abnormal appearance of organism.
 C = Male.
 D = No developing young visible in brood pouch throughout test.
 E = Aborted young.
 F = Young produced but dead at end of the 24 hour period.
 G = Young produced in the brood pouch but not released
 H = Transfer death.

**TOXICITY EVALUATION
Fathead Minnow Summary Table**

Client: City of Cadillac WWTP
Project Number: 211750
Primary Analyst: Michael Amboyan

Conc.	Rep. No.	Day 7 Survivability	Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
0%	1	10	0.41915	0.42163	2.48	0.25
	2	9	0.41009	0.41274	2.65	0.26
		A				
	3	10	0.41445	0.41732	2.87	0.29
	4	9	0.40926	0.41199	2.73	0.27
		A				
5%	5	10	0.41344	0.41624	2.8	0.28
	6	8	0.41174	0.414	2.26	0.23
		A				
	7	10	0.41675	0.41966	2.91	0.29
	8	10	0.41244	0.41513	2.69	0.27
15%	9	10	0.41404	0.41699	2.95	0.3
	10	9	0.41071	0.41311	2.4	0.24
		A				
	11	9	0.40931	0.41162	2.31	0.23
		A				
	12	8	0.41088	0.41357	2.69	0.27
		A				

Qualifier Key:

A = Natural Death
B = Fungal filaments present
C = Lethargy
D = Excessive ventilation
E = Erratic or uncoordinated swimming
F = Transfer death

**TOXICITY EVALUATION
Fathead Minnow Summary Table**

Client: City of Cadillac WWTP

Project Number: 211750

Primary Analyst: Michael Amboyan

30%	13	10	0.41255	0.41538	2.83	0.28
	14	10	0.41628	0.41899	2.71	0.27
	15	9	0.4106	0.41281	2.21	0.22
		A				
	16	6	0.41463	0.41652	1.89	0.19
		A				
60%	17	9	0.40878	0.41168	2.9	0.29
		A				
	18	9	0.42093	0.42337	2.44	0.24
		A				
	19	10	0.41021	0.41339	3.18	0.32
	20	9	0.41474	0.41762	2.88	0.29
		A				
100%	21	9	0.41647	0.42003	3.56	0.36
		A				
	22	9	0.41411	0.41797	3.86	0.39
		A				
	23	10	0.41484	0.41792	3.08	0.31
	24	9	0.40814	0.41122	3.08	0.31
		A				

Qualifier Key:

A = Natural Death

B = Fungal filaments present

C = Lethargy

D = Excessive ventilation

E = Erratic or uncoordinated swimming

F = Transfer death

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: City of Cadillac WWTP

Project Number: 211750

Primary Analyst: Michael Amboyan

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		1	2	3	4	5	6	7	8	9	10			
0%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	2	2	0	0	0	0	0	4	0	8	10	0.8
	4	4	0	0	6	6	4	5	5	0	5	35	10	3.5
	5	10	5	9	10	10	9	10	0	8	12	83	10	8.3
	6	10	16	15	13	12	0	10	0	16	0	92	10	9.2
	7													
	8													
	Total	24	23	26	29	28	13	25	5	28	17	218	10	21.8

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		11	12	13	14	15	16	17	18	19	20			
5%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	5	0	0	0	0	0	0	0	0	5	9	0.6
	4	6	0	5	5	5	6	5	0	5	5	42	9	4.7
	5	11	9	7	12	7	12	10	0	8	8	84	9	9.3
	6	0	16	16	14	16	0	12	0	0	0	74	9	8.2
	7													
	8													
	Total	17	30	28	31	28	18	27	0	13	13	205	9	22.8

*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: City of Cadillac WWTP

Project Number: 211750

Primary Analyst: Michael Amboyan

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		21	22	23	24	25	26	27	28	29	30			
15%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	3	3	0	0	0	0	0	0	0	6	10	0.6
	4	4	0	0	4	6	6	6	5	5	6	42	10	4.2
	5	12	10	10	10	12	11	9	8	13	11	106	10	10.6
	6	11	18	15	0	14	0	14	10	12	0	94	10	9.4
	7													
	8													
	Total	27	31	28	14	32	17	29	23	30	17	248	10	24.8

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		31	32	33	34	35	36	37	38	39	40			
30%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	3	0	0	0	0	0	0	4	0	7	10	0.7
	4	5	0	5	6	6	5	6	7	0	0	40	9	4.4
	5	9	9	9	11	10	10	12	6	10	0	86	9	9.6
	6	0	16	16	14	16	0	9	0	15	0	86	9	9.6
	7													
	8													
	Total	14	28	30	31	32	15	27	13	29	0	219	9	24.3

*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

TOXICITY EVALUATION
Ceriodaphnia Survival and Reproduction Test

Client: City of Cadillac WWTP

Project Number: 211750

Primary Analyst: Michael Amboyan

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		41	42	43	44	45	46	47	48	49	50			
60%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	2	0	0	3	0	0	0	0	0	5	10	0.5
	4	4	0	5	5	6	6	6	6	6	6	50	10	5.0
	5	10	10	6	9	14	9	10	10	12	10	100	10	10.0
	6	10	15	10	12	12	9	12	0	0	0	80	10	8.0
	7													
	8													
	Total	24	27	21	26	35	24	28	16	18	16	235	10	23.5

Concentration	Day	Replicate										No. of Young	No. of Adults	Young per Adult
		51	52	53	54	55	56	57	58	59	60			
100%	0*	1	1	1	1	1	1	1	1	1	1	0	10	0.0
	1	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	2	0	0	0	0	0	0	0	0	0	0	0	10	0.0
	3	0	5	0	0	4	2	0	0	3	0	14	10	1.4
	4	5	0	4	4	0	0	5	4	0	6	28	10	2.8
	5	10	10	8	10	10	9	8	8	11	10	94	10	9.4
	6	13	14	15	16	15	0	0	15	14	0	102	10	10.2
	7													
	8													
	Total	28	29	27	30	29	11	13	27	28	16	238	10	23.8

*Neonates less than 24 hours old added to each replicate vessel at start of test. Values are not included in replicate total.

TOXICITY EVALUATION
Fathead Minnow
Larval Survival and Growth Test

Client: City of Cadillac WWTP

Project Number: 211750

Primary Analyst: Michael Amboyan

Conc.	Rep. No.	Day								Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
		0	1	2	3	4	5	6	7				
0%	1	10	10	10	10	10	10	10	10	0.41915	0.42163	2.48	0.25
	2	10	10	10	10	10	10	9	9	0.41009	0.41274	2.65	0.26
	3	10	10	10	10	10	10	10	10	0.41445	0.41732	2.87	0.29
	4	10	10	10	10	10	10	10	9	0.40926	0.41199	2.73	0.27

Conc.	Rep. No.	Day								Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
		0	1	2	3	4	5	6	7				
5%	5	10	10	10	10	10	10	10	10	0.41344	0.41624	2.8	0.28
	6	10	10	10	9	9	8	8	8	0.41174	0.414	2.26	0.23
	7	10	10	10	10	10	10	10	10	0.41675	0.41966	2.91	0.29
	8	10	10	10	10	10	10	10	10	0.41244	0.41513	2.69	0.27

Conc.	Rep. No.	Day								Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
		0	1	2	3	4	5	6	7				
15%	9	10	10	10	10	10	10	10	10	0.41404	0.41699	2.95	0.3
	10	10	10	10	9	9	9	9	9	0.41071	0.41311	2.4	0.24
	11	10	10	10	10	10	9	9	9	0.40931	0.41162	2.31	0.23
	12	10	10	10	8	8	8	8	8	0.41088	0.41357	2.69	0.27

TOXICITY EVALUATION
Fathead Minnow
Larval Survival and Growth Test

Client: City of Cadillac WWTP

Project Number: 211750

Primary Analyst: Michael Amboyan

Conc.	Rep. No.	Day								Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
		0	1	2	3	4	5	6	7				
30%	13	10	10	10	10	10	10	10	10	0.41255	0.41538	2.83	0.28
	14	10	10	10	10	10	10	10	10	0.41628	0.41899	2.71	0.27
	15	10	10	10	10	10	9	9	9	0.4106	0.41281	2.21	0.22
	16	10	10	10	6	6	6	6	6	0.41463	0.41652	1.89	0.19

Conc.	Rep. No.	Day								Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
		0	1	2	3	4	5	6	7				
60%	17	10	10	10	9	9	9	9	9	0.40878	0.41168	2.9	0.29
	18	10	10	10	10	10	10	9	9	0.42093	0.42337	2.44	0.24
	19	10	10	10	10	10	10	10	10	0.41021	0.41339	3.18	0.32
	20	10	10	10	10	9	9	9	9	0.41474	0.41762	2.88	0.29

Conc.	Rep. No.	Day								Initial Wt. (g)	Final Wt. (g)	Net Wt. (mg)	Avg. Wt. (mg)
		0	1	2	3	4	5	6	7				
100%	21	10	10	10	9	9	9	9	9	0.41647	0.42003	3.56	0.36
	22	10	10	10	9	9	9	9	9	0.41411	0.41797	3.86	0.39
	23	10	10	10	10	10	10	10	10	0.41484	0.41792	3.08	0.31
	24	10	10	10	10	10	10	10	9	0.40814	0.41122	3.08	0.31

TOXICITY EVALUATION
Ceriodaphnia Dubia and Fathead Minnow
Routine Chemical and Physical Determination

Client: City of Cadillac WWTP

Project Number: 211750

Primary Analyst: Michael Amboyan

Parameter	Units	Day							
		0	1	2	3	4	5	6	7

0% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.6	8.7	8.8	7.0	10.3	8.2	8.0	
pH	S.U.	7.8	8.2	8.0	8.0	8.0	7.9	7.7	
Conductivity	umhos/cm	270	280	320	300	290	300	310	

0% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.4	9.0	6.6	9.9	8.0	8.3	0.0	
DO Fathead	mg/L	7.3	7.9	6.3	10.2	7.8	7.1	8.3	
pH Cerio	S.U.	8.2	8.0	7.8	8.0	7.7	7.7	0.0	
pH Fathead	S.U.	7.8	8.0	7.8	8.0	7.8	7.6	8.0	
Conductivity Cerio	umhos/cm	380	310	290	280	320	320	0.0	
Conductivity Fathead	umhos/cm	340	300	300	280	320	320	310	

5% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	8.9	8.7	8.8	6.9	10.1	8.1	8.1	
pH	S.U.	7.7	8.1	7.9	8.0	7.9	7.8	7.6	
Conductivity	umhos/cm	330	600	590	310	320	390	360	

5% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.3	8.8	6.5	9.8	8.0	8.3	0.0	
DO Fathead	mg/L	7.4	8.0	6.3	9.7	7.7	7.0	8.1	
pH Cerio	S.U.	8.2	7.8	7.9	7.9	7.9	7.8	0.0	
pH Fathead	S.U.	7.8	7.9	7.6	7.9	7.6	7.5	7.9	
Conductivity Cerio	umhos/cm	700	660	320	320	380	420	0.0	
Conductivity Fathead	umhos/cm	660	580	350	320	380	400	370	

TOXICITY EVALUATION
Ceriodaphnia Dubia and Fathead Minnow
Routine Chemical and Physical Determination

Client: City of Cadillac WWTP

Project Number: 211750

Primary Analyst: Michael Amboyan

Parameter	Units	Day							
		0	1	2	3	4	5	6	7

15% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	9.0	8.6	8.9	6.9	9.6	8.1	8.1	
pH	S.U.	7.8	8.0	7.6	7.9	7.9	7.7	7.6	
Conductivity	umhos/cm	390	800	780	420	380	460	420	

15% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.4	8.7	6.5	9.8	7.9	8.2	0.0	
DO Fathead	mg/L	7.8	7.9	6.4	9.6	7.6	7.2	8.1	
pH Cerio	S.U.	8.1	7.9	7.7	7.8	8.0	7.9	0.0	
pH Fathead	S.U.	7.8	7.8	7.6	7.8	7.7	7.5	8.0	
Conductivity Cerio	umhos/cm	800	820	430	430	440	480	0.0	
Conductivity Fathead	umhos/cm	760	780	440	450	430	450	430	

30% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	9.0	8.6	8.9	6.9	9.6	8.2	8.2	
pH	S.U.	7.7	8.0	7.5	7.9	8.0	7.7	7.6	
Conductivity	umhos/cm	530	1000	930	500	510	560	530	

30% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.4	8.8	6.5	10.2	8.2	8.2	0.0	
DO Fathead	mg/L	7.8	7.8	6.3	9.6	7.5	7.0	8.1	
pH Cerio	S.U.	8.0	8.0	7.9	7.8	7.8	8.0	0.0	
pH Fathead	S.U.	7.8	7.8	7.6	7.8	7.5	7.5	7.8	
Conductivity Cerio	umhos/cm	1100	1100	530	520	580	600	0.0	
Conductivity Fathead	umhos/cm	1000	930	570	510	550	570	540	

TOXICITY EVALUATION
Ceriodaphnia Dubia and Fathead Minnow
Routine Chemical and Physical Determination

Client: City of Cadillac WWTP

Project Number: 211750

Primary Analyst: Michael Amboyan

Parameter	Units	Day							
		0	1	2	3	4	5	6	7

60% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	9.1	8.9	9.0	6.9	9.3	8.2	8.0	
pH	S.U.	7.7	7.9	7.6	7.9	7.9	7.7	7.6	
Conductivity	umhos/cm	800	1500	1500	810	740	780	710	

60% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.4	8.8	6.4	9.5	7.7	8.2	0.0	
DO Fathead	mg/L	7.4	7.9	6.3	9.4	7.4	6.7	8.0	
pH Cerio	S.U.	8.1	7.9	7.9	7.8	7.8	7.9	0.0	
pH Fathead	S.U.	7.8	7.7	7.6	7.6	7.7	7.4	7.6	
Conductivity Cerio	umhos/cm	1500	1500	820	830	8.0	820	0.0	
Conductivity Fathead	umhos/cm	1500	1500	810	830	780	790	740	

100% - Initial									
Temperature	°C	25	25	25	25	25	25	25	
Dissolved Oxygen (DO)	mg/L	9.1	8.8	8.9	6.8	9.3	8.1	7.9	
pH	S.U.	7.6	7.8	7.5	7.9	7.8	7.7	7.7	
Conductivity	umhos/cm	1100	2100	2000	1100	1000	1100	1000	

100% - Final									
Temperature	°C	25	25	25	25	25	25	25	
DO Cerio	mg/L	8.4	8.8	6.4	9.7	7.6	8.1	0.0	
DO Fathead	mg/L	7.8	7.6	6.2	9.4	7.1	6.5	7.9	
pH Cerio	S.U.	8.3	8.0	8.0	7.8	7.9	8.0	0.0	
pH Fathead	S.U.	7.8	7.6	7.7	7.7	7.6	7.4	7.7	
Conductivity Cerio	umhos/cm	2200	2100	1100	1200	1100	1100	0.0	
Conductivity Fathead	umhos/cm	2100	2000	1100	1100	1100	1100	1100	

Ceriodaphnia Reproduction 211750

Title: Ceriodaphnia Reproduction City of Cadillac WWTP 211750
 File: CR211750.TXT Transform: NO TRANSFORMATION

Kolmogorov Test for Normality

D = 0.1763 (p-value = 0.0001)
 D* = 1.3830

Critical D* = 1.035 (alpha = 0.01 , N = 60)
 = 0.895 (alpha = 0.05 , N = 60)

Data FAIL normality test (alpha = 0.01). Try another transformation.

Warning - The first three homogeneity tests are sensitive to non-normality and should not be performed with this data as is.

Bartlett's Test for Homogeneity of Variance

Calculated B1 statistic = 4.6505 (p-value = 0.4600)

Data PASS B1 homogeneity test at 0.01 level. Continue analysis.

Critical B = 15.0863 (alpha = 0.01, df = 5)
 = 11.0705 (alpha = 0.05, df = 5)

Summary Statistics on Data

TABLE 1 of 2

GRP	IDENTIFICATION	N	MIN	MAX	MEAN
1	Control	10	5.0000	29.0000	21.8000
2	5%	10	0.0000	31.0000	20.5000
3	15%	10	14.0000	32.0000	24.8000
4	30%	10	0.0000	32.0000	21.9000
5	60%	10	16.0000	35.0000	23.5000
6	100%	10	11.0000	30.0000	23.8000

Summary Statistics on Data

TABLE 2 of 2

GRP	IDENTIFICATION	VARIANCE	SD	SEM	C.V. %
1	Control	60.6222	7.7860	2.4622	35.7157
2	5%	100.7222	10.0360	3.1737	48.9563
3	15%	43.5111	6.5963	2.0859	26.5980
4	30%	114.7667	10.7129	3.3877	48.9174
5	60%	35.6111	5.9675	1.8871	25.3936
6	100%	54.4000	7.3756	2.3324	30.9901

Ceriodaphnia Reproduction 211750

ANOVA Table

SOURCE	DF	SS	MS	F
Between	5	125.4833	25.0967	0.3676
Within (Error)	54	3686.7000	68.2722	
Total	59	3812.1833		

(p-value = 0.8686)

Critical F = 3.3769 (alpha = 0.01, df = 5,54)
 = 2.3861 (alpha = 0.05, df = 5,54)

Since $F < \text{Critical } F$ FAIL TO REJECT H_0 : All equal (alpha = 0.01)

Dunnett's Test - TABLE 1 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	TRANSFORMED MEAN	MEAN CALCULATED IN ORIGINAL UNITS	T STAT	SIG 0.01
1	Control	21.8000	21.8000		
2	5%	20.5000	20.5000	0.3518	
3	15%	24.8000	24.8000	-0.8119	
4	30%	21.9000	21.9000	-0.0271	
5	60%	23.5000	23.5000	-0.4601	
6	100%	23.8000	23.8000	-0.5412	

Dunnett critical value = 2.9900 (1 Tailed, alpha = 0.01, df [used] = 5,40)
 (Actual df = 5,54)

Dunnett's Test - TABLE 2 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	NUM OF REPS	MIN SIG DIFF (IN ORIG. UNITS)	% OF CONTROL	DIFFERENCE FROM CONTROL
1	Control	10			
2	5%	10	11.0486	50.7	1.3000
3	15%	10	11.0486	50.7	-3.0000
4	30%	10	11.0486	50.7	-0.1000
5	60%	10	11.0486	50.7	-1.7000
6	100%	10	11.0486	50.7	-2.0000

Steel's Many-One Rank Test - Ho: Control<Treatment

GROUP	IDENTIFICATION	MEAN IN ORIGINAL UNITS	RANK SUM	CRIT. VALUE	DF	SIG 0.01
1	Control	21.8000				
2	5%	20.5000	106.50	67.00	10.00	
3	15%	24.8000	120.00	67.00	10.00	
4	30%	21.9000	115.00	67.00	10.00	
5	60%	23.5000	105.50	67.00	10.00	
6	100%	23.8000	118.50	67.00	10.00	

Critical values are 1 tailed (k = 5)

Ceriodaphnia Reproduction 211750

PMSD >47, therefore the statistics were rerun at alpha level 0.05.

ANOVA Table

SOURCE	DF	SS	MS	F
Between	5	125.4833	25.0967	0.3676
Within (Error)	54	3686.7000	68.2722	
Total	59	3812.1833		

(p-value = 0.8686)

Critical F = 3.3769 (alpha = 0.01, df = 5,54)
= 2.3861 (alpha = 0.05, df = 5,54)

Since F < Critical F FAIL TO REJECT Ho: All equal (alpha = 0.05)

Dunnett's Test - TABLE 1 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	TRANSFORMED MEAN	MEAN CALCULATED IN ORIGINAL UNITS	T STAT	SIG 0.05
1	Control	21.8000	21.8000		
2	5%	20.5000	20.5000	0.3518	
3	15%	24.8000	24.8000	-0.8119	
4	30%	21.9000	21.9000	-0.0271	
5	60%	23.5000	23.5000	-0.4601	
6	100%	23.8000	23.8000	-0.5412	

Dunnett critical value = 2.3100 (1 Tailed, alpha = 0.05, df [used] = 5,40)
(Actual df = 5,54)

Dunnett's Test - TABLE 2 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	NUM OF REPS	MIN SIG DIFF (IN ORIG. UNITS)	% OF CONTROL	DIFFERENCE FROM CONTROL
1	Control	10			
2	5%	10	8.5359	39.2	1.3000
3	15%	10	8.5359	39.2	-3.0000
4	30%	10	8.5359	39.2	-0.1000
5	60%	10	8.5359	39.2	-1.7000
6	100%	10	8.5359	39.2	-2.0000

Steel's Many-One Rank Test - Ho: Control<Treatment

GROUP	IDENTIFICATION	MEAN IN ORIGINAL UNITS	RANK SUM	CRIT. VALUE	DF	SIG 0.05
1	Control	21.8000				
2	5%	20.5000	106.50	75.00	10.00	
3	15%	24.8000	120.00	75.00	10.00	
4	30%	21.9000	115.00	75.00	10.00	
5	60%	23.5000	105.50	75.00	10.00	
6	100%	23.8000	118.50	75.00	10.00	

Critical values are 1 tailed (k = 5)

Fathead Biomass 211750

Title: Fathead Biomass City of Cadillac WWTP 211750
File: fb211750.txt Transform:

NO TRANSFORMATION

Shapiro - Wilk's Test for Normality

D = 1.8756
W = 0.9603

Critical W = 0.8840 (alpha = 0.01 , N = 24)
W = 0.9160 (alpha = 0.05 , N = 24)

Data PASS normality test (alpha = 0.01). Continue analysis.

Bartlett's Test for Homogeneity of Variance

Calculated B1 statistic = 2.5996 (p-value = 0.7614)

Data PASS B1 homogeneity test at 0.01 level. Continue analysis.

Critical B = 15.0863 (alpha = 0.01, df = 5)
= 11.0705 (alpha = 0.05, df = 5)

Summary Statistics on Data

TABLE 1 of 2

GRP	IDENTIFICATION	N	MIN	MAX	MEAN
1	Control	4	2.4800	2.8700	2.6825
2	5%	4	2.2600	2.9100	2.6650
3	15%	4	2.3100	2.9500	2.5875
4	30%	4	1.8900	2.8300	2.4100
5	60%	4	2.4400	3.1800	2.8500
6	100%	4	3.0800	3.8600	3.3950

Summary Statistics on Data

TABLE 2 of 2

GRP	IDENTIFICATION	VARIANCE	SD	SEM	C.V. %
1	Control	0.0265	0.1628	0.0814	6.0676
2	5%	0.0810	0.2845	0.1423	10.6772
3	15%	0.0847	0.2910	0.1455	11.2471
4	30%	0.1923	0.4385	0.2192	18.1943
5	60%	0.0935	0.3057	0.1529	10.7271
6	100%	0.1473	0.3838	0.1919	11.3048

Fathead Biomass 211750

ANOVA Table

SOURCE	DF	SS	MS	F
Between	5	2.3138	0.4628	4.4413
Within (Error)	18	1.8756	0.1042	
Total	23	4.1894		

(p-value = 0.0082)

Critical F = 4.2479 (alpha = 0.01, df = 5,18)
= 2.7729 (alpha = 0.05, df = 5,18)

Since F > Critical F REJECT Ho: All equal (alpha = 0.01)

Title: Fathead Biomass City of Cadillac WWTP 211750

File: fb211750.txt

Transform:

NO TRANSFORMATION

Dunnett's Test - TABLE 1 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	TRANSFORMED MEAN	MEAN CALCULATED IN ORIGINAL UNITS	T STAT	SIG 0.01
1	Control	2.6825	2.6825		
2	5%	2.6650	2.6650	0.0767	
3	15%	2.5875	2.5875	0.4162	
4	30%	2.4100	2.4100	1.1939	
5	60%	2.8500	2.8500	-0.7338	
6	100%	3.3950	3.3950	-3.1216	

Dunnett critical value = 3.2100 (1 Tailed, alpha = 0.01, df = 5,18)

Dunnett's Test - TABLE 2 OF 2 Ho:Control<Treatment

GROUP	IDENTIFICATION	NUM OF REPS	MIN SIG DIFF (IN ORIG. UNITS)	% OF CONTROL	DIFFERENCE FROM CONTROL
1	Control	4			
2	5%	4	0.7327	27.3	0.0175
3	15%	4	0.7327	27.3	0.0950
4	30%	4	0.7327	27.3	0.2725
5	60%	4	0.7327	27.3	-0.1675
6	100%	4	0.7327	27.3	-0.7125



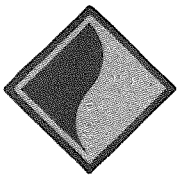
CHAIN-OF-CUSTODY

211750
CADILLAC WWT
City of Cadillac WWTTP

P 734.462.3900 **F** 734.462.3911 **W** www.12649RichfieldCtLivon

Page _____ of _____

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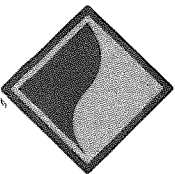
PARAGON
Laboratories

CHAIN-OF-CUSTODY RECORD

12649 Richfield Ct. Livonia, MI 48150
P 734.462.3900 F 734.462.3911 W www.paragonlaboratories.com

Page ____ of ____

Client Name: <u>Cailliac, Kelly</u>				INTERNAL USE ONLY								
Contact Person: <u>Bruce W.</u>				Paragon Report No.: <u>211750</u>								
Mailing Address: <u>3200 W. Lake St.</u>				Logged By: <u>RES</u>								
City, State, Zip: <u>Cailliac, MI 48106</u>				Sample Condition Upon Receipt: <u>Acceptable</u> Other (Specify Below)								
Phone and Fax: <u>(313) 775-3681 / (313) 775-0876</u>				IR Temperature Upon Receipt: <u>4°C</u>								
Email: <u>kelly@cailliac-solutions.com</u>				Remarks: <u>sample received chilled on natural ice - RES 10/18/16</u>								
Client Job Name / No.:												
Job Location:		PO No.:										
Sampled By: <u>FL</u>												
WHERE WAS THE SAMPLE(S) TAKEN: <u>Facility Does Not Chlorinate</u> After Chlorination After Dechlorination				After Chlorination, Before Dechlorination Before Chlorination								
WAS THE SAMPLE(S) DECHLORINATED: YES <u>NO</u> IF YES, PLEASE DESCRIBE DECHLORINATION:												
WAS THE SAMPLE(S) FILTERED: YES <u>NO</u> IF YES, PLEASE STATE THE FILTER MESH SIZE:												
Item No.	Client Sample ID	No. of containers	Grab	No. of Grabs	Grab Sample Period (Start Date and Time - End Date and Time)	Comp	Composite Sample Period (Start Date and Time - End Date and Time)	Ceriodaphnia Dubia	Fathead Minnows	Daphnia Magna	Ceriodaphnia Dubia	Fathead Minnows
1	<u>666666</u>	<u>7</u>				<u>X</u>	<u>10/16/16 7:00 AM - 10/17/16 7:00 AM</u>	<u>X</u>	<u>X</u>		<u>X</u>	<u>X</u>
Tran. #	Released By	Received By	Date	Time	Tran. #	Released By	Received By	Date	Time			
1.	<u>MS vs ms</u>	<u>ECP</u>	<u>10/18/16</u>	<u>9:56</u>	3.							
2.					4.							



PARAGON
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CHAIN-OF-CUSTODY RECORD

12649 Richfield Ct. Livonia, MI 48150
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Page ____ of ____

Client Name: <u>Cadillac Water</u>						INTERNAL USE ONLY					
Contact Person: <u>Amey Val</u>						Paragon Report No.: <u>211756</u>					
Mailing Address: <u>3201 Milgate St</u>						Logged By: <u>RES</u>					
City, State, Zip: <u>Cadillac, MI 49601</u>						Sample Condition Upon Receipt: <u>Acceptable</u> Other (Specify Below)					
Phone and Fax: <u>(231) 775-2368 / (231) 775-2876</u>						IR Temperature Upon Receipt: <u>4°C</u>					
Email: <u>Lab@cadillac-water.com</u>						Remarks: <u>samples received chilled on natural ice - RES 10/19/16</u>					
Client Job Name / No.:											
Job Location:											
Sampled By: <u>TL</u>						PO No.:					
WHERE WAS THE SAMPLE(S) TAKEN: Facility Does Not Chlorinate After Chlorination After Dechlorination Before Chlorination Before Dechlorination											
WAS THE SAMPLE(S) DECHLORINATED: YES <u>NO</u> IF YES, PLEASE DESCRIBE DECHLORINATION:											
WAS THE SAMPLE(S) FILTERED: YES <u>NO</u> IF YES, PLEASE STATE THE FILTER MESH SIZE:											
Item No.	Client Sample ID	No. of containers	Grab	No. of Grabs	Grab Sample Period (Start Date and Time - End Date and Time)	Comp	Composite Sample Period (Start Date and Time - End Date and Time)	ANALYSIS REQUESTED			
1	<u>E44/Grant</u>	<u>2</u>				<u>X</u>	<u>16/12/16 noon - 10/18/16</u>	<u>X</u>	<u>X</u>	<u>X</u>	<u>X</u>
Tran. #	Released By	Received By	Date	Time	Tran. #	Released By	Received By	Date	Time		
1.	<u>MS 3/1/15</u>	<u>EBA</u>	<u>10/19/16</u>	<u>9:50</u>	3.						
2.					4.						